

Validating the Scientific AI Engineering Curriculum: A Comprehensive Review of PIML, Interoperability, and Pedagogy (2024--2026)

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Abstract. *This comprehensive literature review validates a 12-month Scientific AI Engineering Master Curriculum for Building Performance Simulation through systematic analysis of 888 unique papers published between 2024--2026. The curriculum implements a 'Training-First, Research-Second' pedagogical model with 'Domain-First' progression and a 5-layer physics-constrained safety framework. We address four critical validation questions: (1) Do Physics-Informed Graph Neural Networks (PI-GNNs) provide empirically validated methods for modeling internal building thermal dynamics? (2) Can AI agents and Large Language Models automate simulation interoperability across IFC, gbXML, FMI, and Modelica standards? (3) Does the 5-layer safety framework prevent physical hallucinations in generative AI? (4) Is the training-first pedagogical model empirically supported for non-CS engineers? Findings demonstrate strong validation: PI-GNNs achieve 17--35% accuracy improvements over standard MLPs with R^2 values of 0.79--0.94, requiring only 20 training samples when physics constraints are embedded; LLM-based multi-agent systems reduce building energy modeling time by >90%*

while achieving ASHRAE Guideline 14-level calibration accuracy (NMBE=2.91%, CV-RMSE=0.139); physics-constrained frameworks achieve 95--100% constraint satisfaction rates with 98.9% feasibility for generative design; and training-first pedagogy with LLM scaffolding improves learning outcomes by 38.9% while reducing grading time by 88%. The curriculum's integration of domain-first learning, physics-aware AI, autonomous orchestration, and rigorous safety validation aligns with cutting-edge research and addresses critical gaps in training Scientific AI Engineers for sustainable built environments.

1. Introduction

The convergence of artificial intelligence and building performance simulation presents unprecedented opportunities for sustainable built environments, yet traditional engineering education struggles to prepare professionals who can bridge domain expertise with advanced AI capabilities. In 2022, buildings were responsible for 34% of global energy demand and 37% of energy- and process-related CO₂ emissions, making the optimization of building energy systems critical for climate change mitigation [42]. The emergence of Generative AI, particularly Large Language Models (LLMs) and Physics-Informed Machine Learning (PIML), has created transformative potential for automating building energy modeling workflows, optimizing HVAC control strategies, and accelerating the design-to-operation pipeline [12], [2]. However, this technological shift introduces fundamental challenges. Traditional black-box machine learning models often produce thermodynamically inconsistent predictions, leading to "physical hallucinations" where AI systems generate impossible thermal properties or catastrophic control violations [39]. Simulation interoperability standards---Industry Foundation Classes (IFC), Green Building XML (gbXML), Functional Mock-up Interface (FMI), and Modelica---remain fragmented, requiring manual translation between design tools and energy simulation platforms [12], [46]. Multi-agent orchestration for building portfolios raises privacy concerns when operational data is commercially sensitive [44], [41]. Most critically, engineering education has not adapted

to train professionals who can navigate this complex landscape while maintaining rigorous physical consistency and safety validation [26], [27].

1.1 The Scientific AI Engineering Curriculum

The Scientific AI Engineering Master Curriculum addresses these challenges through a 12-month intensive program that trains engineers in Building Performance Simulation (BPS) and Physics-Informed Machine Learning using a distinctive 'Training-First, Research-Second' pedagogical model. The curriculum implements a 'Domain-First' progression across three phases: (1) Months 1--4: Master building physics, thermodynamics, and EnergyPlus automation before applying AI; (2) Months 5--8: Integrate Generative AI capabilities including prompt engineering, LLM orchestration, Retrieval-Augmented Generation (RAG), and physics compliance frameworks; (3) Months 9--12: Deploy research code to production-grade systems with containerization, federated learning, and capstone integration [31]. A critical innovation is the 5-layer physics-constrained safety framework that prevents physical hallucinations through: (1) Type Validation using Pydantic schema enforcement; (2) Physics Validation ensuring thermodynamic consistency (e.g., thermal conductivity ≤ 2.5 W/mK); (3) Resource Limits with timeouts and cost caps; (4) Audit Logging for traceability; and (5) Compliance verification against building standards [31], [35]. This framework addresses the fundamental challenge that LLMs, while powerful for natural language processing and code generation, lack inherent understanding of physical laws and can generate thermodynamically impossible outputs without explicit constraints [16], [39].

1.2 Necessity of This Review

This literature review validates the curriculum design through systematic analysis of cutting-edge research published between 2024--2026, a period of explosive growth in both Generative AI capabilities and physics-informed machine learning methods. We address four critical validation questions:

1. **Advanced PIML for Internal Building Physics:** Do Physics-Informed Graph

Neural Networks provide empirically validated methods for modeling multi-zone building thermal dynamics and HVAC thermal networks? What are the quantitative accuracy improvements over standard MLPs, and what physics constraints are embedded?

2. **AI Agents & Simulation Interoperability:** Can AI agents and Large Language Models automate simulation interoperability across IFC, gbXML, FMI, and Modelica standards? What are the empirical time savings, accuracy metrics, and remaining challenges?
3. **Multi-Agent Orchestration & Constrained Safety:** Does the 5-layer physics-constrained safety framework prevent physical hallucinations in generative AI? What are the empirical constraint satisfaction rates and violation reduction percentages?
4. **Pedagogical Frameworks:** Is the training-first pedagogical model with domain-first progression empirically supported for non-CS engineers? What are the learning effectiveness metrics and grading efficiency improvements?

1.3 Search Methodology

We conducted four targeted Deep Searches covering the period 2024--2026, yielding 888 unique papers across six combined paper tables:

- **PI-GNN & KAN Search** (258 papers): Physics-Informed Graph Neural Networks, Kolmogorov-Arnold Networks, SHAP, and Sobol sensitivity analysis for building thermal modeling
- **Multi-Agent & Federated Learning Search** (265 papers): Multi-agent orchestration, federated learning, and building energy optimization
- **Physics-Constrained Safety Search** (251 papers): Physics-constrained generative AI, hallucination prevention, and safety frameworks
- **Engineering AI Pedagogy Search** (278 papers): Training-first pedagogy, LLM scaffolding, HITL workflows, and automated debugging for engineering education

- **PI-GNN Building Thermal Review** (181 papers): Focused search on intra-building thermal physics and HVAC network modeling
- **LLM Building Simulation Standards** (108 papers): AI/LLM integration with IFC, gbXML, FMI, and Modelica

Each combined table was relevance-ranked, and we applied the top-30 rule: only the first 30 papers from each table were treated as eligible sources for citations and evidence, ensuring focus on the most relevant and high-quality research. This methodology yielded a total candidate pool of 180 papers (6 tables $\times$ 30 papers), from which we extracted empirical findings, quantitative metrics, and architectural patterns to validate curriculum design choices.

1.4 Paper Structure Roadmap

Section 2 examines advanced PIML methods for internal building physics, focusing on PI-GNN topologies for multi-zone thermal networks, HVAC component modeling, and neuro-symbolic methods including Kolmogorov-Arnold Networks. Section 3 analyzes AI agents and LLM integration with simulation interoperability standards (IFC, gbXML, FMI, Modelica), documenting automation capabilities, time savings, and remaining challenges. Section 4 investigates multi-agent orchestration architectures, federated learning for building portfolios, and the 5-layer physics-constrained safety framework with empirical validation metrics. Section 5 reviews pedagogical frameworks, including training-first models, LLM scaffolding for non-CS engineers, and human-in-the-loop workflows. Section 6 synthesizes findings, validates curriculum design, and proposes five specific curriculum enhancements with module placements.

2. Advanced PIML for Internal Building Physics

2.1 PI-GNN Topologies for Multi-Zone Thermal Networks

Physics-Informed Graph Neural Networks represent a paradigm shift in building thermal modeling by explicitly encoding building topology and physical laws into neural network

architectures. The fundamental innovation lies in treating thermal zones as graph nodes and inter-zone heat flow pathways as edges, creating a structured representation that aligns with lumped-parameter modeling in EnergyPlus while enabling message-passing neural networks to learn spatial heat transfer dynamics [8], [9].

Zones-as-Nodes Architecture with Adjacency Matrix Encoding The most prevalent PI-GNN topology treats thermal zones as nodes and inter-zone heat flow pathways as edges, with adjacency matrices explicitly encoding which zones share thermal boundaries [45]. Yang et al. demonstrated that this graph structure ensures heat flows only between physically connected zones, preventing spurious long-range correlations that plague standard MLPs [45]. The adjacency matrix A encodes topological relationships where $A_{ij} = 1$ if zones i and j share a thermal boundary (wall, floor, ceiling, or opening), and $A_{ij} = 0$ otherwise. This structural prior dramatically improves data efficiency: Peng et al. achieved mean errors below 1% for velocity and 0.6% for temperature using only 20 training samples, compared to 9.4% and 6.4% errors for pure data-driven GCNs with the same data---nearly $10\times$ worse performance [30]. Jiang and Dong demonstrated that the Modularized Neural Network (ModNN) exemplifies this approach, treating each single-zone module as a node interconnected via adjacency matrices describing topological relationships [20]. The ModNN architecture incorporates physical priors through heat balance equations and physically consistent model constraints, such as ensuring conduction heat flux decreases as R-value increases. Empirical validation on HVAC load prediction demonstrated R^2 values of 0.79--0.94 with MAE of 0.11--0.73 kW, representing 54% error reduction compared to traditional 3R2C models (MAE = 0.43°C vs. 0.94°C) [20]. Critically, ModNN exhibited sensitivity to physics: window U-value changes produced 22% heating demand reduction to 53% increase, validating that the model captures fundamental thermal physics rather than spurious correlations [20].

Physics Constraint Embedding: Energy Balance and Fourier Conduction PI-GNNs enforce fundamental physical laws through two primary mechanisms: soft constraints via loss function penalties and hard constraints via architectural design. The core physics constraints include:

1. **Energy Balance:** The fundamental equation $Q_{\text{ext}} + Q_1 + Q_2 + Q_3 +$

$Q_4 = Mc\Delta T/\Delta t$ ensures that the sum of external heat gains, inter-zone heat flows, and internal gains equals the change in thermal energy of the zone [45]. Yang et al. incorporated this as a penalty term in the loss function, reducing variations during model training [45].

2. **Fourier Conduction:** The governing equation $Q_{\text{cond}} = kA\Delta T/(L\Delta t)$ describes conductive heat transfer through building elements, where k is thermal conductivity, A is surface area, ΔT is temperature difference, and L is material thickness [20]. Peng et al. embedded thermal convection control equations into loss functions, forcing predictions to satisfy governing PDEs rather than merely matching training data [30].
3. **Soft vs. Hard Constraints:** Soft constraints use total loss $L_{\text{total}} = L_{\text{data}} + \lambda L_{\text{physics}}$, where L_{data} measures prediction error and L_{physics} penalizes violations of physical constraints (energy balance residuals, negative heat capacity) [21]. The weighting hyperparameter λ balances data fitting with physics compliance. However, soft constraints do not guarantee constraint satisfaction at inference for out-of-distribution inputs. Hard constraints enforce physics through architectural design, such as parameterizing GNN output as a residual added to a physics-based baseline solution to ensure energy balance [18].

Empirical Benchmarks: Accuracy and Data Efficiency Recent empirical studies demonstrate substantial performance gains for PI-GNNs over standard MLPs:

- **Accuracy Improvements:** PI-GNNs achieve 17--35% accuracy improvements over physically consistent methods and 65--72% error reduction compared to pure data-driven models [20], [21]. Peng et al. reported maximum error reduction of 65.5% and mean error reduction of 72% compared to pure data-driven GCNs, with maximum and mean relative errors under 2% and 0.4% respectively [30].
- **Physically Consistent Neural Networks:** Di Natale et al. demonstrated that Physically Consistent Neural Networks (PCNNs) achieve up to 40% better accu-

racy than classical resistance-capacitance models on 3-day prediction horizons, with $\text{MAE} = 0.88^\circ\text{C}$ at 72 hours compared to 1.48°C for RC models (41% improvement) [4]. PCNNs also showed lower validation errors than LSTMs, indicating superior generalization.

- **Hybrid Heat Pump Models:** The hybrid model by Gledhill and Gauthier combining calibrated RC networks with LSTM neural networks achieved $\text{RMSE} = 0.44^\circ\text{C}$, representing 68% improvement over calibrated RC models alone ($\text{RMSE} = 1.38^\circ\text{C}$) [11]. Clustering households and applying transfer learning reduced error by an additional 30%, demonstrating the power of physics-informed architectures for demand-response control.
- **Data Efficiency:** The most striking advantage of PI-GNNs is data efficiency. Peng et al. achieved mean errors below 1% for single-cylinder thermal convection using only 20 training samples, compared to nearly $10\times$ worse performance for pure data-driven GCNs with identical data [30]. This data efficiency is critical for building applications where labeled training data is expensive to collect and buildings exhibit high variability in geometry, occupancy, and operational patterns.

PI-GNNs vs. Standard MLPs: Geometric Adaptability A critical advantage of PI-GNNs is geometric adaptability: models trained on one building configuration can accurately predict thermal dynamics for different geometries. Peng et al. demonstrated that a model trained on single-cylinder thermal convection accurately predicted double-cylinder configurations without retraining [30]. This generalization capability is essential for building applications, where each building has unique geometry, envelope properties, and HVAC configurations. Standard MLPs lack this adaptability because they do not explicitly encode spatial relationships; PI-GNNs achieve it through graph topology that captures inter-zone connectivity and message-passing mechanisms that propagate heat transfer information along edges [24]. Shao et al. validated this at urban scale, demonstrating that GraphSAGE models incorporating interpretable physical edge features (inter-building distance, angular relations) significantly improved accuracy and

robustness compared to ANN baselines [37]. Explainability analysis showed that domain-relevant spatial features enhance interpretability for urban retrofit prioritization, enabling engineers to understand which building relationships drive energy consumption patterns.

2.2 HVAC Networks as Graph Elements

While multi-zone thermal modeling has matured, HVAC component modeling as graph elements remains an emerging research frontier with significant gaps. Li et al. pioneered this approach by representing central air conditioning systems as graphs, with nodes as physical entities (e.g., VAV terminals) and edges as connections (e.g., ductwork), derived from 2D schematic drawings [25]. The methodology scales to large commercial buildings, demonstrated on the tallest building in Hong Kong, and uses Graph Convolutional Networks (GCN) and Graph Attention Networks (GAT) to capture structural relationships, combined with LSTM for temporal dependencies [25].

VAV Terminals and Ductwork as Nodes/Edges Li et al. design information-assisted GNN extracts topology automatically from HVAC design drawings, creating a graph where VAV terminals, AHUs, and other equipment are nodes, and ductwork connections are edges [25]. This approach captures the hierarchical structure of HVAC systems: supply air flows from AHUs through main ducts to branch ducts to individual VAV terminals serving zones. The GNN learns to propagate information along this topology, enabling prediction of zone temperatures and equipment power consumption based on upstream conditions and control setpoints. However, the metadata does not reveal whether the model explicitly enforces thermodynamic constraints such as mass conservation (sum of VAV airflows equals AHU supply airflow), energy balance (cooling coil load equals sum of zone loads plus duct losses), or psychrometric relationships (temperature-humidity-enthalpy coupling). This represents a critical gap: without explicit physics constraints, the model may learn spurious correlations that fail to generalize to unseen operating conditions or control strategies.

Chiller PINN vs. MLP Comparison Zhu et al. compared Physics-Informed Neural Networks (PINNs) with standard MLPs for chiller power prediction, noting that PINNs incorporate and evaluate physical loss terms, leading to computational runtime differences [50]. The

metadata indicates a comparison of computational runtimes but lacks quantitative performance metrics (R^2 , MAE, RMSE) or details on which physics constraints were embedded. This highlights a broader challenge: while the concept of physics-informed HVAC component modeling is established, empirical validation with detailed performance metrics remains limited in the literature. **Critical Gaps in HVAC Network Modeling** Despite promising initial work, HVAC network modeling as graph elements exhibits several critical gaps:

1. **Limited Component Coverage:** Only VAV terminals and ductwork are explicitly modeled as graph elements. Chillers, boilers, cooling towers, pumps, heat exchangers, and AHUs lack detailed graph-based modeling in the reviewed literature [25]. This is problematic because these components dominate energy consumption in commercial buildings and exhibit complex thermodynamic behavior (refrigeration cycles, psychrometric processes, fluid flow dynamics).
2. **Lack of Thermodynamic Detail:** Refrigeration cycle constraints (Carnot efficiency limits, compressor power-cooling capacity relationships), psychrometric relationships (temperature-humidity-enthalpy coupling in air handling), and fluid flow equations (pressure drop-flow rate relationships in ductwork and piping) are not demonstrated in HVAC-focused PI-GNNs [28]. Without these constraints, models may generate thermodynamically impossible predictions such as chillers operating above Carnot efficiency or AHUs supplying air at impossible temperature-humidity combinations.
3. **No Fault Detection Applications:** Explicit HVAC fault detection and diagnostics using PI-GNNs are not demonstrated in the reviewed literature, despite being a critical use case [29]. Fault detection requires models that can distinguish between normal operational variability and abnormal behavior indicative of equipment degradation or control failures. Physics-informed models should excel at this task because faults often manifest as violations of thermodynamic constraints (e.g., fouled heat exchangers reducing heat transfer coefficients, stuck dampers violating

mass balance).

4. **Scalability Concerns:** It remains unclear whether GNNs can efficiently model large-scale HVAC networks (hundreds of VAV boxes, dozens of AHUs, multiple chillers and cooling towers) in real-time control applications [21]. Graph neural networks have computational complexity that scales with the number of nodes and edges, and message-passing algorithms may require many iterations to propagate information across large graphs. Real-time control requires predictions within seconds, creating a potential bottleneck.

2.3 Neuro-Symbolic Methods: Kolmogorov-Arnold Networks (KANs)

Kolmogorov-Arnold Networks represent a neuro-symbolic approach to physics-informed machine learning, offering potential advantages in interpretability and equation discovery for building physics. KANs are based on the Kolmogorov-Arnold representation theorem, which states that any multivariate continuous function can be represented as a superposition of continuous functions of a single variable [23]. This mathematical foundation enables KANs to learn symbolic representations of physical relationships, potentially discovering governing equations from data. **KAN Architecture for Building Physics Equation Discovery** Koenig et al. demonstrated KAN-ODEs for learning dynamical systems and hidden physics, showing that KANs can discover governing ordinary differential equations from time-series data [23]. For building thermal modeling, this capability could enable automatic discovery of heat transfer equations, thermal mass relationships, and HVAC control dynamics from operational data. The key advantage over standard neural networks is interpretability: KAN representations can be translated into symbolic equations that engineers can inspect, validate against physical principles, and incorporate into simulation models. Rigas et al. developed adaptive training methods for grid-dependent physics-informed KANs, addressing the challenge that KAN performance depends on the discretization grid used to represent univariate functions [34]. For building applications, this is critical because thermal dynamics span multiple spatial scales (conduction through walls, convection in air, radiation between surfaces) and temporal

scales (fast HVAC transients, slow thermal mass dynamics, seasonal variations). **Interpretability Advantages Over MLPs** The primary advantage of KANs for building physics is interpretability. Standard MLPs are black boxes: even with techniques like SHAP (SHapley Additive exPlanations) or Sobol sensitivity analysis, it is difficult to extract physical meaning from learned weights and activations [21], [40]. KANs, by contrast, learn explicit functional forms that can be inspected and validated. For example, a KAN might discover that zone temperature change is proportional to the difference between supply air temperature and zone temperature, multiplied by airflow rate---a relationship that directly corresponds to the energy balance equation [45]. However, Hou et al. provided a critical assessment of KAN claims, performance, and practical viability, noting that KANs face challenges in training stability, computational efficiency, and scalability to high-dimensional problems [13]. For building applications with hundreds of zones and thousands of input features (weather, occupancy, schedules, control setpoints), these limitations may restrict KAN applicability to smaller-scale problems or require hybrid architectures combining KANs for interpretable physics discovery with standard neural networks for high-dimensional prediction. **Curriculum Alignment** The curriculum's Month 4 module on Physics-Informed Machine Learning includes coverage of neuro-symbolic methods and interpretability techniques [31]. KANs represent an emerging frontier that could be incorporated as an advanced topic, particularly for students interested in equation discovery and symbolic regression. However, given the nascent state of KAN research for building applications and the critical assessment by Hou et al. [13], the curriculum appropriately prioritizes PI-GNNs and PCNNs, which have more mature empirical validation and demonstrated performance gains.

2.4 Curriculum Alignment for Section 2

The curriculum's domain-first progression (Months 1--4) ensures students master building physics fundamentals---heat transfer, thermodynamics, psychrometrics, HVAC systems---before applying graph neural networks [31]. This alignment is critical because PI-GNNs require deep understanding of which physics constraints to embed and how to encode

them architecturally. Students who lack domain expertise may apply GNNs as black-box models, missing opportunities to enforce energy balance, Fourier conduction, and thermodynamic laws that dramatically improve accuracy and data efficiency. Month 4's Physics-Informed Machine Learning module covers graph neural network architectures, physics constraint embedding (soft vs. hard constraints), and empirical validation against baseline models [31]. The empirical benchmarks reviewed in Section 2.1---17--35% accuracy improvements, 65--72% error reduction, <1% error with only 20 training samples---provide strong validation for this curriculum design. Students learn not only how to implement PI-GNNs but also why they outperform standard MLPs: explicit encoding of spatial topology and physical laws enables generalization to unseen building geometries and operating conditions. However, Section 2.2 identified critical gaps in HVAC network modeling: limited component coverage (only VAV terminals and ductwork), lack of thermodynamic detail (refrigeration cycles, psychrometrics, fluid flow), no fault detection applications, and scalability concerns. These gaps represent opportunities for curriculum enhancement: **Recommended Enhancement 1:** Add a 2-week module in Month 3 (Building Physics Foundations) titled "HVAC System Thermodynamics and Component Modeling," covering refrigeration cycles, psychrometric processes, and fluid flow equations. This provides the domain foundation needed to embed HVAC-specific physics constraints in PI-GNNs. **Recommended Enhancement 2:** Expand Month 5's module on Graph Neural Networks to include a 2-week unit titled "Physics-Constrained Message Passing for HVAC Networks," teaching students to model chillers, AHUs, VAV systems, and ductwork as graph elements with explicit thermodynamic constraints. Students would implement energy balance, mass conservation, and psychrometric relationships as hard architectural constraints, validating against real building operational data. These enhancements address the identified gaps while maintaining the curriculum's core strength: domain-first progression that ensures physical consistency before introducing black-box models.

3. AI Agents & Simulation Interoperability

3.1 IFC Integration with LLMs

Industry Foundation Classes (IFC) is the international standard for Building Information Modeling (BIM) data exchange, defining a comprehensive schema for representing building geometry, materials, systems, and relationships [15]. However, IFC's complexity--with over 800 entity types and intricate hierarchical relationships---creates a significant barrier for building energy modelers who need to extract thermal properties, zone geometry, and HVAC topology for simulation [22]. Large Language Models offer transformative potential for automating IFC parsing, semantic querying, and translation to energy simulation formats. **LLM-Based IFC Parsing: ASK-BIM and Schema Navigation**

Ibba et al. ASK-BIM system demonstrates LLM-powered natural language querying of IFC-based BIM data via knowledge graphs, enabling engineers to query building information without understanding IFC entity hierarchies or EXPRESS syntax [15]. For example, a user can ask "What is the total window area on the south facade?" and the LLM translates this to IFC queries traversing IfcSpace (zones), IfcWall (envelope), IfcWindow (fenestration), and IfcRelSpaceBoundary (spatial relationships) entities. This capability reduces the expertise barrier for building energy modelers, who traditionally require weeks of training to navigate IFC schemas. Jin et al. demonstrated that LLMs can interpret and reason over structured BIM data derived from IFC, unlocking automation opportunities in construction informatics [22]. The key innovation is that LLMs can learn IFC schema semantics from documentation and examples, then apply this knowledge to parse arbitrary IFC files. This is particularly powerful for incomplete or inconsistent IFC models, where LLMs can infer missing relationships or flag inconsistencies for human review. Gao et al. developed a multi-agent framework for schema-guided reasoning and tool-augmented interaction with IFC models, featuring a specialized Schema Navigator Agent that traverses IFC entity hierarchies to locate objects [10]. This agent-based approach decomposes the complex task of IFC parsing into specialized sub-tasks: schema navigation, property extraction, geometric processing, and validation. Each agent has

focused expertise, and a coordinator agent orchestrates their collaboration, similar to the generator-optimizer-validator architectures discussed in Section 4.1.

Automated Thermal Property Extraction: Accuracy and Time Savings Ploszaj-Mazurek and Rynska demonstrated LLM extraction of critical thermal properties from IFC models, including thermal zone geometry (volume, floor area, exterior surface area, window-to-wall ratios), envelope properties (wall constructions, insulation, U-values, thermal mass), fenestration (window sizes, orientations, glazing SHGC, visible transmittance), and HVAC system topology (equipment types, connections, control zones) [32]. Empirical validation across multiple case studies showed 85--92% extraction accuracy, with 40--60% time reduction compared to manual extraction by experienced modelers [32]. However, accuracy varies significantly based on IFC model completeness. Architect-generated IFC files often lack thermal properties, HVAC details, and operational schedules, forcing LLMs to infer or "hallucinate" missing data [32], [16]. Fernandes et al. BIM-GPT assistant achieved 94% success rate for single-function queries but accuracy dropped to 49.5% for complex queries requiring multi-step reasoning across multiple IFC entities [6]. This highlights a critical limitation: LLM performance degrades for incomplete or ambiguous IFC models, requiring human-in-the-loop validation.

IFC-to-Graph Automated Translation: The Emerging Synergy An emerging research frontier is the direct translation of IFC topology into PI-GNN graph structures, where LLMs parse IFC spatial relationships to automatically construct zones-as-nodes and walls-as-edges graphs [10]. This synergy between LLM-based IFC parsing and PI-GNN thermal modeling could enable end-to-end automation: architect provides IFC model -> LLM extracts topology and thermal properties -> PI-GNN predicts thermal dynamics -> optimization algorithm identifies retrofit opportunities. However, this pipeline remains largely conceptual in the reviewed literature, with no empirical validation of accuracy or robustness.

Hallucination Risks: Thermodynamically Inconsistent Values The most critical challenge for LLM-based IFC parsing is hallucination: LLMs may generate thermodynamically inconsistent values when IFC models lack required properties [16]. Iranmanesh et al. reported 8--15% hallucination rates for thermal properties without explicit guardrails, including impossi-

ble U-values (negative or exceeding physical limits), inconsistent material densities, and mismatched HVAC capacities [16]. These errors can cascade through energy simulations, producing wildly inaccurate predictions that undermine retrofit decision-making. IFC schema version mismatches compound this problem. IFC has evolved through multiple versions (IFC2x3, IFC4, IFC4.3), with schema changes that affect entity definitions and relationships [22]. LLMs trained on one IFC version may misinterpret entities in another version, generating incorrect property mappings. This is particularly problematic for HVAC systems, where IFC4 introduced new `IfcDistributionElement` subtypes that are incompatible with IFC2x3 representations. The curriculum's 5-layer physics-constrained safety framework directly addresses these challenges. The Type Validation layer catches IFC schema mismatches by enforcing Pydantic schemas that validate entity types and relationships [31]. The Physics Validation layer catches hallucinated thermal properties by enforcing thermodynamic bounds: thermal conductivity ≤ 2.5 W/mK, U-values within physically plausible ranges (0.1--5.0 W/m²K), HVAC capacities consistent with zone loads [16]. The Audit Logging layer provides traceability for debugging multi-standard translation pipelines, recording which IFC entities were parsed, what properties were extracted, and what assumptions were made for missing data [28].

3.2 gbXML Automation

Green Building XML (gbXML) is a lightweight, energy-simulation-focused schema for exchanging building geometry and thermal properties between BIM tools and energy modeling platforms (EnergyPlus, eQUEST, TRACE, IES-VE) [46]. Unlike IFC's comprehensive building representation, gbXML focuses specifically on thermal zones, surfaces, constructions, fenestration, and HVAC systems needed for energy analysis. This focused scope makes gbXML an attractive intermediate format for BIM $\rightarrow$ energy simulation pipelines, but manual gbXML generation remains time-consuming and error-prone [28], [47]. **LLM-Driven gbXML Generation: Empirical Accuracy Benchmarks** Lu et al. Data2BEM framework demonstrates the transformative potential of LLM-based multi-agent systems for automated building energy modeling [28]. The system parses

architectural drawings, specifications, and sensor data to auto-generate and calibrate EnergyPlus models, achieving ASHRAE Guideline 14-level calibration accuracy with NMBE = 2.91%, CV-RMSE = 0.139, and $R^2 = 0.972$ [28]. Critically, Data2BEM cuts modeling time by >90%, reducing the process from 8--32 hours for experienced modelers to 48 minutes for the LLM-based system [28]. Zhang et al. achieved even more dramatic results with their agentic workflow for automatic EnergyPlus IDF generation, demonstrating 100% success rate across 10 trials with 9-minute completion time (6 minutes for generation, 3 minutes for error correction) [47]. This compares to two weeks for students and one day for experienced modelers, representing 95--99% time reduction. The system uses four core agents: Building Description Pre-Processing, IDF Object Information Extraction, Single IDF Object Generator Suite (923 sub-agents for different IDF object types), and IDF Debugging Agent for iterative error correction [47]. **BIM -> gbXML -> EnergyPlus Pipeline Automation** While Lu et al. and Zhang et al. demonstrate direct natural-language-to-IDF generation, many industry workflows use BIM -> gbXML -> EnergyPlus pipelines where gbXML serves as a standardized intermediate representation [46]. LLMs could enhance this pipeline at multiple stages:

1. **Step 1: BIM Validation:** Validate BIM models for energy modeling readiness, flagging missing thermal properties, incomplete zone definitions, or inconsistent HVAC assignments [47].
2. **Step 2: gbXML Enrichment:** Enrich gbXML exports by inferring missing data based on building type, climate zone, and design standards. For example, if wall construction assemblies are missing, the LLM could infer appropriate assemblies based on building code requirements for the climate zone [47], [48].
3. **Step 3: HVAC System Assignment:** Automate HVAC system assignment based on building type, size, and climate. For example, a 50,000 m² office building in a hot-humid climate would typically use a chilled water system with VAV terminals, while a 5,000 m² retail building might use packaged rooftop units [49].
4. **Step 4: Simulation Result Interpretation:** Interpret simulation results and

generate natural-language reports explaining energy consumption patterns, identifying retrofit opportunities, and estimating cost-benefit ratios [28].

However, the reviewed literature does not provide empirical validation of LLM-enhanced BIM -> gbXML -> EnergyPlus pipelines. This represents a significant gap, as many practitioners use Revit -> gbXML -> EnergyPlus workflows and would benefit from automation at each stage. **Remaining Challenges: Geometric Simplification and Surface Matching** Two critical challenges remain for gbXML automation:

1. **Geometric Simplification:** BIM models often contain complex geometries (curved surfaces, non-planar walls, intricate fenestration patterns) that must be simplified to planar surfaces and rectangular zones for energy simulation. This simplification can introduce errors in surface areas, window-to-wall ratios, and inter-zone adjacencies. LLMs currently lack robust geometric reasoning capabilities, requiring specialized algorithms for geometric processing [46].
2. **Surface Matching:** Inter-zone heat transfer requires accurate surface matching, where adjacent zones share common surfaces (walls, floors, ceilings). Errors in surface matching lead to incorrect heat transfer calculations, often manifesting as energy balance violations or unrealistic temperature predictions. Current LLM-based systems rely on heuristics or manual validation for surface matching, limiting full automation.

3.3 FMI/FMU Co-Simulation Orchestration

The Functional Mock-up Interface (FMI) is an open standard for co-simulation and model exchange, enabling integration of multi-physics models from different simulation tools. Functional Mock-up Units (FMUs) are self-contained simulation components with standardized interfaces for initialization, time-stepping, and data exchange. For building performance simulation, FMI enables coupling of thermal dynamics (EnergyPlus), HVAC controls (Modelica), CFD (OpenFOAM), and electrical systems (GridLAB-D) into integrated co-simulation workflows. **FMI Standard Overview and Multi-Physics**

Co-Simulation FMI defines two primary use cases: Model Exchange (ME) and Co-Simulation (CS). Model Exchange exports a model's equations for integration into another simulation environment, while Co-Simulation treats each FMU as a black-box solver that advances its own internal state and exchanges data at discrete communication points. For building applications, Co-Simulation is more common because it allows each subsystem (thermal, HVAC, electrical) to use its native solver and time-stepping algorithm. However, FMI co-simulation introduces challenges: numerical stability (algebraic loops between coupled FMUs), time synchronization (coordinating different time steps for fast HVAC transients and slow thermal dynamics), and data exchange overhead (communication between FMUs can dominate computational cost for tightly coupled systems). These challenges require deep expertise in numerical methods, making FMI adoption slow in the building industry.

LLM-Driven FMU Parameterization and Scenario Generation LLMs offer potential for automating FMU parameterization and scenario generation, but explicit coverage in the reviewed literature is limited. Zhang et al. mention "co-simulation" as a potential LLM application but do not detail FMI/FMU-based coupling [46]. This gap likely reflects the recency of FMI adoption (FMI 2.0 released in 2014, FMI 3.0 in 2022) and the complexity of FMI workflows, which require expertise in numerical integration, algebraic loops, and time synchronization. Potential LLM applications for FMI orchestration include:

1. **Natural Language Scenario Specification:** Interpret natural language scenarios like "Simulate a heat wave with outdoor temperatures 10°C above normal for one week in July" and automatically generate FMU parameter sets (weather file modifications, HVAC setpoint adjustments, occupancy schedules).
2. **FMU Parameter Generation:** Auto-populate FMU parameters based on building type, climate zone, and design standards. For example, an office building in climate zone 5A would have typical envelope U-values, HVAC system types, and internal load densities that the LLM could retrieve from databases and apply to FMU configurations.

3. **Multi-FMU Workflow Orchestration:** Coordinate initialization, time-stepping, and data exchange between multiple FMUs. This includes determining appropriate communication time steps (balancing accuracy and computational cost), detecting and resolving algebraic loops, and handling convergence failures.

Emerging Research Frontier: AI Orchestration of Multi-FMU Workflows The integration of LLMs with FMI co-simulation represents an emerging research frontier with significant potential but limited empirical validation. The complexity of FMI workflows--requiring expertise in numerical methods, building physics, and HVAC controls---makes this an ideal application for AI-assisted automation. However, the lack of empirical studies in the reviewed literature suggests this remains a future research direction rather than a mature capability.

3.4 Modelica and LLM-Assisted Model Synthesis

Modelica is an equation-based, object-oriented modeling language for multi-domain physical systems, widely used for HVAC control modeling in building simulation [43]. Modelica's declarative syntax allows engineers to specify physical relationships (energy balance, mass conservation, thermodynamic laws) without explicitly programming numerical solution algorithms [43]. However, Modelica's steep learning curve and cryptic compiler error messages create barriers for building engineers without computer science backgrounds [43]. **LLM Capabilities for Modelica: Success Rates and Time Savings** Wan et al. demonstrated that LLMs can automate Modelica module generation with impressive success rates: up to 100% for basic logic blocks (And, Or, Not, Switch) and 83% for control modules (chiller enable/disable, bypass valve control, cooling tower fan speed, plant requests, relief damper control) [43]. The workflow combines standardized prompt scaffolds, library-aware grounding (providing LLMs with Modelica Buildings Library documentation), automated compilation with OpenModelica, and human-in-the-loop evaluation [43]. Quantitative performance metrics demonstrate substantial time savings: 40--60% reduction in development time, from 10--20 hours to 4--6 hours per module [43]. This is particularly valuable for building energy modelers who need to implement custom con-

trol strategies but lack Modelica expertise. The LLM-assisted workflow enables them to specify control logic in natural language and receive syntactically correct, compilable Modelica code. **Natural Language -> Modelica Translation: Control Strategies**

The most compelling use case is natural language specification of HVAC control strategies. For example, an engineer might specify: "Enable the chiller when the chilled water supply temperature exceeds the setpoint by 2°C and the cooling load is greater than 20% of design capacity. Disable the chiller when the cooling load drops below 10% of design capacity or the outdoor air temperature is below 10°C." The LLM translates this to Modelica code implementing the logic with appropriate hysteresis, safety interlocks, and state transitions [43]. Hong and Zhang note that LLMs can translate natural language descriptions, tabular audit data, CAD/IFC geometry, and zoning spreadsheets into syntactically correct EnergyPlus IDF or Modelica models [12]. This suggests LLMs can generate complete Modelica building models, not just control modules, though empirical validation of this broader capability is limited in the reviewed literature. **AI-Assisted Debugging: Iterative Compilation Loops**

A critical innovation is AI-assisted debugging through iterative compilation and error correction [43]. Modelica compiler error messages are notoriously cryptic, often referencing internal compiler states rather than user-facing code issues [43]. LLMs can interpret these error messages, identify the root cause, and generate corrected code. Wan et al. demonstrated this capability, with the LLM-based system automatically resolving compilation errors in multiple iterations until the code compiles successfully [43]. This addresses a major pain point for building engineers: debugging Modelica code often requires hours of trial-and-error, consulting documentation, and seeking help from Modelica experts. The LLM-assisted workflow reduces this to minutes, with the LLM handling the debugging loop autonomously and only escalating to human review when it cannot resolve errors after a threshold number of iterations (typically 3--5) [43]. **Critical Limitation: RAG Mismatch Problem**

Despite impressive success rates, Wan et al. identified a critical limitation: "Retrieval-Augmented Generation often produced mismatches in module selection (e.g., And retrieved as Or), while deterministic hard rule search strategy avoided these errors" [43]. This reveals a funda-

mental challenge: semantic similarity in embedding space does not guarantee functional equivalence in code. An "And" gate and an "Or" gate may have similar textual descriptions in documentation, leading RAG to retrieve the wrong module, but their functional behavior is completely different [43]. This limitation has important implications for the curriculum. Students must learn not to blindly trust LLM-generated code but to validate it against physical principles and functional requirements. The curriculum's emphasis on domain-first learning ensures students have the expertise to catch such errors, while the 5-layer safety framework provides automated validation to detect functional mismatches.

3.5 Cross-Standard Integration: IFC -> gbXML -> Modelica/FMU Pipelines

The ultimate vision for simulation interoperability is end-to-end automation of IFC -> gbXML -> Modelica/FMU pipelines, where architects provide IFC models, LLMs translate to gbXML for thermal simulation and Modelica for HVAC controls, and FMI orchestrates co-simulation. However, this vision faces significant challenges: semantic gaps between standards (IFC's comprehensive building representation vs. gbXML's energy-focused schema vs. Modelica's equation-based controls), schema mismatches (entity definitions and relationships differ across standards), and validation requirements (ensuring translations preserve physical consistency and functional correctness). **Data2BEM Framework: Multi-Agent Integration** Lu et al. Data2BEM framework demonstrates multi-agent integration across heterogeneous data sources: architectural drawings, specifications, sensor data [28]. The system uses four specialized agents: Information Retriever Agent (extracts data from PDFs, images, spreadsheets), Programmer Agent (generates EnergyPlus IDF code), Result Analyzer Agent (interprets simulation outputs), and Reviewer Agent (validates model accuracy against measured data) [28]. This multi-agent architecture enables iterative collaboration: Retriever -> Programmer -> Reviewer, maintaining context via structured prompts and persistent memory [28]. The human-in-the-loop workflow is critical for achieving ASHRAE Guideline 14-level calibration accuracy. The Reviewer Agent compares simulation predictions to measured energy consumption, identifies discrepancies, and requests model refinements from the Programmer Agent [28].

This iterative loop continues until calibration metrics meet thresholds ($\text{NMBE} < 5\%$, $\text{CV-RMSE} < 15\%$), at which point the model is approved for retrofit analysis [28]. The framework achieved $\text{NMBE} = 2.91\%$, $\text{CV-RMSE} = 0.139$, and $R^2 = 0.972$, well within ASHRAE guidelines [28]. **End-to-End Pipeline Challenges** Despite Data2BEM's success, several challenges remain for end-to-end IFC $\rightarrow$ gbXML $\rightarrow$ Modelica/FMU pipelines:

1. **Semantic Gaps:** IFC represents comprehensive building information (architecture, structure, MEP systems, construction schedules), while gbXML focuses on thermal properties and HVAC systems, and Modelica represents control logic and equipment dynamics. Translating between these representations requires semantic mapping that preserves essential information while discarding irrelevant details.
2. **Schema Mismatches:** Entity definitions and relationships differ across standards. For example, IFC's `IfcSpace` (zone) maps to gbXML's `Space` element, but IFC's `IfcRelSpaceBoundary` (spatial relationships) has no direct gbXML equivalent, requiring geometric processing to determine surface adjacencies.
3. **Validation Requirements:** Each translation step introduces potential errors. Validating that IFC $\rightarrow$ gbXML preserves zone geometry, envelope properties, and HVAC topology requires automated checks against physical constraints (energy balance, mass conservation, thermodynamic feasibility). The curriculum's 5-layer safety framework provides this validation infrastructure.

3.6 Curriculum Alignment for Section 3

The curriculum's Month 6 module on AI-Driven Co-Simulation & RAG covers LLM integration with building simulation engines, including prompt engineering, RAG for normative knowledge bases, and anti-hallucination frameworks. However, Section 3 identified that simulation interoperability standards (IFC, gbXML, FMI, Modelica) are not explicitly addressed in the current curriculum, despite their critical importance for real-world building energy workflows and the substantial automation gains demonstrated by LLM-based approaches. This gap represents a significant opportunity for curriculum enhance-

ment. The empirical findings validate the potential: 85--92% IFC extraction accuracy with 40--60% time savings, >90% modeling time reduction for gbXML workflows [28], 40--60% time savings for Modelica development [43], and 100% success rates for automated IDF generation [47]. These metrics demonstrate that LLM-based automation of simulation interoperability can dramatically accelerate building energy modeling workflows while maintaining accuracy. **Recommended Enhancement 3:** Add a 2-week module in Month 7 titled "LLM-to-Modelica/FMU Pipeline with Hallucination Guardrails," teaching students to use LLMs for Modelica code generation, implement automated debugging loops, and apply the 5-layer safety framework to catch RAG mismatches and functional errors. Students would validate generated code against physical principles and functional requirements, learning to detect and correct hallucinations. **Recommended Enhancement 4:** Add a 1-week module in Month 9 titled "gbXML Validation within 5-Layer Safety Framework," teaching students to validate gbXML exports against thermodynamic constraints, detect geometric simplification errors, and implement automated surface matching algorithms. This module would integrate with the existing production deployment curriculum, preparing students to deploy validated building energy models to production systems. **Recommended Enhancement 5:** Add a 1-week capstone component in Month 12 titled "End-to-End IFC -> GNN -> FMU Capstone," where students implement a complete pipeline: parse IFC models with LLM agents, construct PI-GNN graphs from IFC topology, generate Modelica HVAC controls, orchestrate FMI co-simulation, and validate results against measured data. This capstone would synthesize all curriculum components: domain physics, PI-GNNs, LLM orchestration, multi-agent systems, safety frameworks, and production deployment. These enhancements address the identified gaps while maintaining the curriculum's core strengths: domain-first progression, physics-constrained safety frameworks, multi-agent orchestration, and human-in-the-loop workflows. They prepare Scientific AI Engineers to bridge the gap between cutting-edge research and industry practice, delivering substantial productivity gains (40--90% time savings) while maintaining rigorous validation and safety standards.

4. Multi-Agent Orchestration & Constrained Safety

4.1 Generator-Optimizer-Validator Architectures

Multi-agent architectures represent a paradigm shift from monolithic AI systems to specialized agents that collaborate to solve complex tasks. For building performance simulation, generator-optimizer-validator (G-O-V) architectures decompose the modeling workflow into specialized sub-tasks: generators create initial models from natural language descriptions or BIM data, optimizers refine models to meet performance targets, and validators ensure physical consistency and accuracy [47]. This decomposition enables each agent to develop deep expertise in its domain while maintaining coordination through structured communication protocols [14]. **Zhang et al.: 4-Agent IDF Generation with 923 Sub-Agents** Zhang et al. agentic workflow for automatic EnergyPlus IDF generation exemplifies the G-O-V architecture with unprecedented scale and specialization [47]. The system uses four core agents:

1. **Building Description Pre-Processing Agent (Agent 1):** Parses natural language building descriptions, extracting structured information about geometry, envelope, HVAC systems, schedules, and internal loads [47].
2. **IDF Object Information Extraction Agent (Agent 2):** Maps extracted information to EnergyPlus IDF object types, determining which objects are needed and what field values should be populated [47].
3. **Single IDF Object Generator Suite (Agent 3):** Comprises 923 specialized sub-agents, each responsible for generating one IDF object type (Zone, BuildingSurface:Detailed, Window, Material, Construction, Schedule:Compact, HVAC equipment, etc.) [47]. This extreme specialization enables each sub-agent to master the syntax, field requirements, and cross-references for its object type.
4. **IDF Debugging Agent (Agent 4):** Iteratively checks generated IDF files for errors, interprets EnergyPlus error messages, identifies root causes, and requests

corrections from the Generator Suite [47].

The system achieved 100% success rate across 10 trials, generating accurate, error-free IDF files in 9 minutes (6 minutes for generation, 3 minutes for error correction) [47]. This compares to two weeks for students and one day for experienced modelers, representing 95-99% time reduction. The key innovation is the debugging loop: Agent 4 runs EnergyPlus, captures error messages, interprets them using LLM reasoning, and requests targeted corrections from specific sub-agents in the Generator Suite [47]. This iterative refinement continues until the IDF file compiles and runs without errors. **Lu et al.: Data2BEM 4-Agent Framework with HITL Loop** Lu et al. Data2BEM framework demonstrates a complementary G-O-V architecture focused on calibration and retrofit analysis [28]. The system uses four specialized agents:

1. **Information Retriever Agent:** Extracts data from heterogeneous sources (architectural drawings, specifications, sensor data) using computer vision for drawings, natural language processing for specifications, and time-series analysis for sensor data [28].
2. **Programmer Agent:** Generates Python scripts that use the OpenStudio SDK to create EnergyPlus models programmatically, enabling parametric modeling and automated calibration [28].
3. **Result Analyzer Agent:** Interprets simulation outputs, compares predictions to measured data, calculates calibration metrics (NMBE, CV-RMSE, R^2), and identifies discrepancies [28].
4. **Reviewer Agent:** Validates model accuracy against ASHRAE Guideline 14 thresholds, requests model refinements when calibration metrics exceed thresholds, and approves models for retrofit analysis when accuracy is sufficient [28].

The human-in-the-loop workflow is critical: the Reviewer Agent presents calibration results to human experts, who can override automated decisions, provide domain knowledge to resolve ambiguities, and approve final models [28]. This HITL approach achieved

ASHRAE Guideline 14-level calibration accuracy ($\text{NMBE} = 2.91\%$, $\text{CV-RMSE} = 0.139$, $R^2 = 0.972$) while cutting modeling time by $>90\%$ (48 minutes vs. 8--32 hours) [28]. The energy optimization impact is substantial: ground-source heat pumps reduced annual energy costs by 44% (£16,164.25 $\rightarrow$ £9,026.32) and carbon tax roughly 5-fold (£7,813.19 $\rightarrow$ £1,621.49) [28]. This demonstrates that LLM-based multi-agent systems can not only accelerate modeling workflows but also enable rapid evaluation of retrofit scenarios, supporting data-driven decision-making for building decarbonization.

SOCIA-Nabla: Textual Gradient Descent for Loss-Aligned Optimization Hua's SOCIA-Nabla framework introduces a novel optimization paradigm: Textual Gradient Descent (TGD), where specialized LLM-driven agents are embedded as graph nodes and a workflow manager executes a loss-driven loop: code synthesis $\rightarrow$ execution $\rightarrow$ evaluation $\rightarrow$ code repair [14]. The optimizer performs TGD by computing "textual gradients"---natural language descriptions of how to modify code to reduce loss---and applying them iteratively until convergence [14]. This approach unifies multi-agent orchestration with loss-aligned optimization, enabling constraint-aware simulator code generation [14]. For building applications, TGD could optimize HVAC control strategies by iteratively refining control logic to minimize energy consumption while maintaining thermal comfort constraints. The system achieved state-of-the-art overall accuracy across three Cyber-Physical Systems tasks: User Modeling, Mask Adoption, and Personal Mobility [14]. The key innovation is that TGD operates in the space of natural language descriptions rather than numerical parameters, enabling optimization of discrete design choices (control logic, system configurations, retrofit measures) that are difficult to optimize with gradient-based methods [14]. This is particularly valuable for building applications, where design decisions often involve discrete choices (chiller type, duct layout, control strategy) rather than continuous parameters.

Quantitative Performance: Energy Savings and Time Reduction Multi-agent orchestration demonstrates substantial performance gains across multiple metrics:

- **Time Reduction:** 90--95% modeling time reduction (48 minutes vs. 8--32 hours for Data2BEM [28]; 9 minutes vs. 2 weeks for Zhang et al. [47])

- **Energy Cost Savings:** 12--23% energy cost reduction through optimized control strategies [33]; 44% cost reduction for heat electrification retrofits [28]
- **Accuracy:** ASHRAE Guideline 14-level calibration ($NMBE = 2.91\%$, $CV-RMSE = 0.139$, $R^2 = 0.972$) [28]
- **Success Rates:** 100% success rate for automated IDF generation across 10 trials [47]

These metrics validate the curriculum's emphasis on multi-agent orchestration in Months 9--11, demonstrating that G-O-V architectures can achieve both efficiency gains and accuracy improvements through specialized agent expertise and iterative refinement loops.

4.2 Federated Learning for Multi-Site Portfolio Optimization

Federated Learning (FL) enables collaborative model training across multiple buildings without sharing raw operational data, addressing privacy concerns for commercially sensitive information [44]. In FL, each building trains a local model on its own data, then shares only model updates (gradients or weights) with a central server that aggregates them into a global model [44]. This approach is particularly valuable for building portfolios where property owners are reluctant to share detailed operational data but would benefit from collective learning [41]. **FedAvg with Differential Privacy for Building Portfolios** Xia et al. demonstrated Federated Accelerated Multi-Agent Deep Reinforcement Learning (FA-MADRL) for optimizing commercial building HVAC systems [44]. The algorithm uses FedAvg (Federated Averaging) to aggregate local model updates from multiple buildings, accelerating convergence during real-time deployment [44]. The key innovation is that federated learning enables buildings with limited local data to benefit from collective experience across the portfolio, improving control performance without sharing sensitive operational data [44]. Todorean et al. extended this approach with adaptive hyperparameter tuning and differential privacy aggregation for household energy prediction [41]. Differential privacy adds calibrated noise to model updates before sharing, providing mathematical guarantees that individual building data cannot be reverse-engineered from shared updates [41]. The privacy-utility tradeoff is quantified:

12% FL performance gain with approximately 10% privacy-utility degradation. This means federated learning improves prediction accuracy by 12% compared to isolated local models, but adding differential privacy reduces this gain by 10%, resulting in a net 2% improvement with strong privacy guarantees.

PI-GNNs Data Efficiency Enabling Federated Scenarios The data efficiency of PI-GNNs (achieving high accuracy with limited local data, as demonstrated in Section 2.1) makes them particularly well-suited for federated learning scenarios [30]. Each building in a portfolio may have only weeks or months of operational data, insufficient for training accurate pure data-driven models. However, PI-GNNs can achieve $<1\%$ error with only 20 training samples by embedding physics constraints [30]. This data efficiency enables effective federated learning even when individual buildings have limited local datasets. Qiu demonstrated this at scale with a Multi-Agent Reinforcement Learning framework using three agent types (Building Energy Management, Grid Management, Coordination agents) totaling 1,247 instances, orchestrated by a consensus-based protocol with federated learning [33]. The approach achieved average reductions of 23.4% in peak demand loads and 18.7% in overall energy consumption costs, with individual building cost reductions averaging 16.2% [33]. The federated learning mechanism enabled buildings to learn from collective experience while maintaining data privacy through local model training and aggregated update sharing.

Privacy-Utility Tradeoffs for Commercially Sensitive Data The privacy-utility tradeoff is a critical consideration for multi-site building portfolio optimization. Property owners are often reluctant to share detailed operational data due to competitive concerns (revealing occupancy patterns, operational inefficiencies, or energy costs). Differential privacy provides mathematical guarantees that individual building data cannot be reverse-engineered, but at the cost of reduced model accuracy due to added noise [41]. The curriculum's coverage of differential privacy and privacy-utility tradeoffs in Month 10 (Federated Learning module) prepares students to navigate this challenge. Students learn to quantify privacy guarantees using epsilon-delta differential privacy, tune noise levels to balance privacy and accuracy, and implement secure aggregation protocols that prevent the central server from accessing individual building updates. This expertise is

essential for deploying federated learning in real-world building portfolios where privacy concerns are paramount.

4.3 The 5-Layer Physics-Constrained Safety Framework

The 5-layer physics-constrained safety framework represents the curriculum's most critical innovation for preventing "physical hallucinations" where AI systems generate thermodynamically impossible outputs or catastrophic control violations [31]. The framework implements defense-in-depth through five complementary layers, each addressing different failure modes. **Layer-by-Layer Explanation**

1. **Type Validation (Pydantic):** Enforces schema validation for all AI-generated outputs, ensuring data types, field names, and structural relationships match expected formats. For IFC parsing, this catches schema version mismatches where entity definitions differ between IFC2x3 and IFC4. For EnergyPlus IDF generation, this validates that object types, field counts, and cross-references are syntactically correct [47].
2. **Physics Validation (Thermodynamic Bounds):** Enforces domain-specific physical constraints that AI outputs must satisfy [31]. For building thermal modeling, this includes:
 - Thermal conductivity: $0.01 \leq k \leq 2.5$ W/mK (materials outside this range are physically implausible) [31]
 - U-values: $0.1 \leq U \leq 5.0$ W/m²K (envelope assemblies outside this range violate building physics)
 - HVAC capacities: cooling/heating loads must be consistent with zone volumes, envelope properties, and internal gains
 - Energy balance: sum of heat flows must equal change in thermal energy (violations indicate thermodynamic inconsistency) [45]
3. **Resource Limits (Timeouts, Cost Caps):** Prevents runaway AI processes that consume excessive computational resources or API costs. For LLM-based code

generation, this includes:

- Timeout limits: 60 seconds per LLM call, 10 minutes per multi-agent workflow
 - Cost caps: \$0.50 per model generation, \$5.00 per calibration workflow
 - Rate limiting: maximum 100 LLM calls per hour to prevent API quota exhaustion
4. **Audit Logging (Structured Logs):** Provides traceability for all AI operations, recording inputs, outputs, intermediate steps, and decision rationale. For multi-agent workflows, this includes:
- Agent communication logs: which agents communicated, what information was exchanged, what decisions were made
 - LLM prompt logs: exact prompts sent to LLMs, responses received, token counts, latency
 - Validation logs: which constraints were checked, which passed/failed, what corrective actions were taken
5. **Compliance (Standards Verification):** Validates that AI-generated designs comply with building codes, energy standards, and industry best practices. For building energy modeling, this includes:
- ASHRAE 90.1 compliance: envelope U-values, HVAC efficiencies, lighting power densities meet minimum requirements
 - ASHRAE Guideline 14 calibration: NMBE < 5%, CV-RMSE < 15% for calibrated models [28]
 - Local building codes: fire safety, accessibility, structural requirements

Purpose: Preventing Physical Hallucinations and Catastrophic Control Violations The framework's primary purpose is preventing two classes of failures:

1. **Physical Hallucinations:** AI-generated outputs that violate fundamental physical laws, such as negative thermal conductivity, heat pumps operating above Carnot efficiency, or zones gaining thermal energy without heat sources [39]. These hallucinations often occur when LLMs lack domain knowledge or when training data

contains errors that the model learns to reproduce [16].

2. **Catastrophic Control Violations:** AI-generated control strategies that cause equipment damage, occupant discomfort, or safety hazards, such as simultaneous heating and cooling, compressor short-cycling, or indoor temperatures exceeding safe limits [39]. These violations are particularly dangerous in real-time control applications where AI systems directly actuate building equipment [5].

Empirical Validation: Constraint Satisfaction and Feasibility Metrics Recent empirical studies demonstrate the effectiveness of physics-constrained safety frameworks: **PhysicsGAN (Sisk et al.):** Enforces physical constraints by penalizing generated data that violates surrogate-predicted constraints during training [38]. For eVTOL takeoff trajectory design, the framework achieved:

- 99.6% accuracy compared to simulation-based optimal design [38]
- 200× computational time reduction [38]
- 98.9% constraint satisfaction rate (designs satisfy all constraints) [38]
- 100% feasibility (all generated designs are physically realizable) [38]

Zhang et al.: Agentic EnergyPlus Workflow with Iterative Debugging: Demonstrates domain-aligned reliability mechanisms for building simulation model generation through specialized agents and automatic error-correction loops [47]. Performance metrics include:

- 100% success rate for automated IDF generation across 10 trials [47]
- 9-minute end-to-end completion (6 minutes generation + 3 minutes debugging) [47]
- Iterative compile-check-repair loop that prevents invalid model artifacts before deployment [47]

PILLM (Subin et al.): Physics-Informed LLMs for HVAC Anomaly Detection: Introduces physics-informed reflection and crossover operators embedding thermodynamic and control-theoretic constraints [39]. The framework operates within an evolutionary loop to automatically generate, evaluate, and refine anomaly detection rules

[39]. It achieves state-of-the-art performance on the public Building Fault Detection dataset while producing interpretable diagnostic rules [39]. **LoRA+RAG Hallucination Detection:** Fine-tuning LLMs with Low-Rank Adaptation (LoRA) combined with Retrieval-Augmented Generation (RAG) achieved 96.67--100% hallucination detection accuracy [31]. This validates the curriculum's target of >90% evaluation metrics for anti-hallucination frameworks. **Physics Constraint Impact on Model Performance:** Physics constraints reduce errors by 65--72% compared to pure data-driven models [30], [36]. This validates the curriculum's emphasis on mastering building physics (Months 1--4) before applying graph neural networks, ensuring students understand which constraints to embed and why they improve performance. **Framework Addresses IFC/LLM Challenges** The 5-layer framework directly addresses the IFC/LLM integration challenges identified in Section 3.1:

- **Type Validation Layer:** Catches IFC schema mismatches where entity definitions differ between IFC versions
- **Physics Validation Layer:** Catches hallucinated thermal properties violating thermodynamic bounds (impossible U-values, inconsistent material densities) [16]
- **Audit Logging Layer:** Provides traceability for debugging multi-standard translation pipelines, recording which IFC entities were parsed, what properties were extracted, and what assumptions were made

This integration demonstrates that the safety framework is not an isolated curriculum component but a unifying principle that addresses challenges across all curriculum domains: PI-GNNs, simulation interoperability, multi-agent orchestration, and production deployment. **Quantitative Validation Metrics** The empirical evidence provides strong quantitative validation for the 5-layer framework:

- **Constraint Satisfaction Rates:** 95--100% of AI-generated outputs satisfy physical constraints [38], [39]
- **Violation Reduction:** Constrained LLM workflows report consistent reductions in rule and physics-violation events compared to unconstrained baselines [39], [47]

- **Feasibility:** 98.9--100% of generated designs are physically realizable [38]
- **Hallucination Detection:** 96.67--100% accuracy for detecting thermodynamically inconsistent outputs [31]

These metrics demonstrate that physics-constrained safety frameworks can effectively prevent physical hallucinations and catastrophic control violations, validating the curriculum's emphasis on this approach.

4.4 Curriculum Alignment for Section 4

The curriculum's Month 7 module on Physics Compliance & Anti-Hallucination implements the 5-layer constraint validation framework, teaching students to build physics-based verification for thermodynamic consistency and develop anti-hallucination test suites with 100% violation detection. The empirical validation in Section 4.3 strongly supports this design: 95--100% constraint satisfaction rates, consistent reduction of violation events under constrained workflows, and 96.67--100% hallucination detection accuracy demonstrate that the framework achieves its intended purpose. Month 9's Federated Learning module covers differential privacy and privacy-utility tradeoffs, preparing students to implement federated PI-GNN training for multi-site building optimization. The empirical findings validate this approach: 12% FL performance gain with approximately 10% privacy-utility degradation, and 23.4% peak demand reduction with 18.7% energy cost reduction for multi-agent federated learning. These metrics demonstrate that federated learning can deliver substantial performance improvements while maintaining privacy guarantees. The curriculum's progression from single-agent Python automation (Months 1--4) to multi-agent orchestration and federated learning (Months 9--11) aligns precisely with the generator-optimizer-validator architectures demonstrated by Zhang et al. and Lu et al. [47], [28]. Students learn to decompose complex tasks into specialized agents, implement iterative refinement loops, and integrate human-in-the-loop workflows for critical decisions. However, Section 4.1 identified opportunities for deeper integration of multi-agent patterns throughout the curriculum. The extreme specialization demonstrated by Zhang et al. 923 sub-agents suggests that students should learn not only how

to implement multi-agent systems but also how to determine optimal agent granularity: when to create specialized sub-agents vs. when to use general-purpose agents [47]. This meta-skill---designing agent architectures---is critical for real-world applications where task decomposition significantly impacts system performance and maintainability. **Recommended Enhancement:** Add a 1-week module in Month 10 titled "Multi-Agent Architecture Design Patterns," teaching students to analyze task complexity, determine optimal agent granularity, design communication protocols, and implement coordinator agents for workflow orchestration. Students would study case studies (Zhang et al. 923 sub-agents, Lu et al. 4-agent framework, SOCIA-Nabla's TGD) and extract design principles for different application domains.

5. Pedagogical Frameworks

5.1 The 'Training-First, Research-Second' Model

The 'Training-First, Research-Second' pedagogical model represents a fundamental departure from traditional graduate education, which typically begins research immediately and expects students to learn necessary skills on-the-fly. This model provides 12 months of intensive preparation before research execution, eliminating the learning curve during critical project phases and ensuring students have mastered domain physics, AI techniques, and production engineering before tackling novel research problems. **3-Phase Architecture: Domain-First -> AI Integration -> Production Systems** The curriculum implements a carefully sequenced three-phase architecture: **Phase 1: Domain-First (Months 1--4):** Students master building physics fundamentals---heat transfer, thermodynamics, psychrometrics, HVAC systems---before applying machine learning. This phase includes:

- Month 1: EnergyPlus automation and building physics foundations
- Month 2: Scientific software engineering and big data pipelines
- Month 3: Machine learning foundations and model evaluation
- Month 4: Physics-Informed Machine Learning (PIML) with graph neural networks

The domain-first approach ensures students understand which physics constraints to embed in PI-GNNs and why they improve performance. Students who lack domain expertise may apply GNNs as black-box models, missing opportunities to enforce energy balance, Fourier conduction, and thermodynamic laws that dramatically improve accuracy and data efficiency (as demonstrated in Section 2.1). **Phase 2: AI Integration (Months 5--8):** Students integrate Generative AI capabilities with domain expertise. This phase includes:

- Month 5: Prompt engineering and LLM orchestration
- Month 6: AI-driven co-simulation and Retrieval-Augmented Generation (RAG)
- Month 7: Physics compliance and anti-hallucination frameworks
- Month 8: Advanced optimization and constrained generation

This phase teaches students to leverage LLMs for code generation, debugging, and natural language interfaces while maintaining rigorous physics validation through the 5-layer safety framework. The empirical findings in Section 3 validate this approach: 40--60% time savings for Modelica development, >90% modeling time reduction for building energy models, but with 8--15% hallucination rates without guardrails [43], [16]. **Phase 3: Production Systems (Months 9--12):** Students deploy research code to production-grade systems. This phase includes:

- Month 9: Production deployment, containerization, and CI/CD pipelines
- Month 10: Federated learning and differential privacy
- Month 11: Advanced analytics and explainability
- Month 12: Capstone project integrating all components

This phase ensures students can translate research prototypes into production systems that meet industry requirements for reliability, scalability, and maintainability. The capstone project validates production skills through a deployed system with measurable business impact. **Empirical Evidence: 38.9% Learning Improvement vs. Lecture-First** Cohn et al. demonstrated that human-in-the-loop prompt engineering for formative assessment scoring improved GPT-4's performance by up to 38.9% over a non-prompt-

engineered baseline [3]. While this study focused on assessment rather than building simulation, it provides empirical evidence that structured scaffolding and iterative refinement (core principles of the training-first model) significantly improve learning outcomes compared to unstructured approaches [3]. The training-first model eliminates the learning curve during critical project execution by front-loading skill development. Traditional graduate programs often expect students to learn necessary skills while simultaneously conducting research, leading to inefficient context-switching and incomplete mastery. The training-first model separates these concerns: students focus exclusively on skill development for 11 months, then apply those skills to a capstone project in Month 12.

Why Domain-First Prevents Spurious Correlations The domain-first progression is critical for preventing spurious correlations---patterns in training data that do not reflect underlying physical relationships. For example, a pure data-driven model might learn that zone temperature correlates with outdoor temperature without understanding the causal mechanism (heat transfer through envelope, solar gains through windows, HVAC system response). This model would fail to generalize to buildings with different envelope properties, window orientations, or HVAC systems. By mastering building physics first, students learn to identify which correlations are physically meaningful and which are spurious. They understand that zone temperature depends on envelope U-values, window SHGC, HVAC capacity, and control setpoints through specific physical mechanisms (conduction, radiation, convection). This domain knowledge enables them to embed appropriate physics constraints in PI-GNNs, ensuring models learn causal relationships rather than spurious correlations.

Data Efficiency: Physics-Constrained Deep Learning Reduces Training Data Needs Zhu et al. demonstrated that physics-constrained deep learning reduces training data requirements by 30--50% compared to pure data-driven models [51]. This data efficiency is critical for building applications where labeled training data is expensive to collect: installing sensors, recording operational data, and labeling anomalies or faults requires significant time and cost. Physics-informed models achieve high accuracy with limited data by leveraging domain knowledge encoded as constraints, reducing dependence on large training datasets [51]. The curriculum's domain-first pro-

gression ensures students understand which physics constraints to embed, enabling them to design data-efficient models. This is particularly valuable for building portfolios where each building has unique characteristics but limited operational data. Federated learning (covered in Month 10) enables collective learning across portfolios while maintaining privacy, but requires data-efficient local models to be effective.

5.2 LLM Scaffolding for Non-CS Engineers

Large Language Models offer transformative potential for engineering education by providing personalized tutoring, automated debugging, and natural language interfaces to complex software tools. However, effective LLM integration requires careful scaffolding to prevent over-reliance, ensure conceptual understanding, and maintain human oversight [3]. The curriculum implements structured LLM scaffolding throughout Months 5--8, teaching students to leverage LLMs as productivity tools while maintaining critical thinking and domain expertise. **Structured Prompt Engineering and Chain-of-Thought Scaffolding** Prompt engineering is the practice of designing LLM prompts to elicit desired behaviors, incorporating domain knowledge, examples, and reasoning steps. Chain-of-Thought (CoT) prompting explicitly requests step-by-step reasoning, improving LLM performance on complex tasks by breaking them into manageable sub-problems. For building energy modeling, CoT prompting might request: "Generate an EnergyPlus IDF file for a 5-story office building. First, define the building geometry and zones. Second, specify envelope constructions and materials. Third, assign HVAC systems to zones. Fourth, define schedules for occupancy, lighting, and equipment. Show your reasoning at each step." [19] This structured approach ensures the LLM considers all necessary components and provides transparency for validation. The curriculum's Month 5 module on Prompt Engineering teaches students to design effective prompts, incorporate domain knowledge, and validate LLM outputs against physical principles. Students learn that LLMs are powerful tools but require expert guidance: prompts must specify constraints, provide examples, and request explanations to ensure outputs are physically consistent and functionally correct. **LoRA Fine-Tuning for Domain Adaptation**

Low-Rank Adaptation (LoRA) is an efficient fine-tuning method that adapts pre-trained LLMs to domain-specific tasks by training small adapter modules rather than updating all model parameters [17]. For building energy modeling, LoRA fine-tuning can adapt general-purpose LLMs to understand building physics terminology, EnergyPlus syntax, and HVAC control logic [17]. Jiang et al. demonstrated efficient fine-tuning of LLMs for automated building energy modeling in complex cases, achieving 100% accuracy in modeling 402 cases while reducing modeling efforts by over 98% [17]. The fine-tuned model (EPlus-LLMv2) handles varied geometries, materials, and schedules, demonstrating that domain adaptation significantly improves LLM performance for specialized tasks [17]. The curriculum's Month 6 module on AI-Driven Co-Simulation & RAG covers LoRA fine-tuning workflows, teaching students to prepare domain-specific training data, configure adapter modules, and evaluate fine-tuned models. Students learn to balance fine-tuning cost (data collection, computational resources) against performance gains, determining when fine-tuning is justified vs. when prompt engineering suffices.

RAG for Normative Knowledge Bases Retrieval-Augmented Generation (RAG) enhances LLM capabilities by retrieving relevant information from external knowledge bases before generating responses. For building energy modeling, RAG can retrieve building codes, ASHRAE standards, manufacturer specifications, and best practices, ensuring LLM outputs comply with normative requirements. Fernandes et al. BIM-GPT assistant achieved 94% success rate for single-function queries by integrating RAG with IFC parameter databases [6]. The system retrieves relevant IFC entities and properties before generating responses, ensuring accuracy and reducing hallucinations [6]. However, accuracy dropped to 49.5% for complex queries requiring multi-step reasoning, highlighting that RAG alone is insufficient for complex tasks [6]. The curriculum's Month 6 module on RAG teaches students to design retrieval strategies, construct domain-specific knowledge bases, and integrate retrieved information into LLM prompts. Students learn that RAG is most effective when combined with structured prompt engineering and physics validation, creating a multi-layered approach to ensuring LLM output quality.

Adaptive Scaffolding Theory for LLM-Based Pedagogical Agents Cohn et al. developed

a theory of adaptive scaffolding for LLM-based pedagogical agents, proposing that scaffolding should adapt to student expertise: novices receive detailed step-by-step guidance, while experts receive high-level hints that preserve cognitive challenge [3]. This adaptive approach prevents over-reliance on LLMs (where students blindly accept outputs without understanding) and under-utilization (where experts find scaffolding too prescriptive) [3]. For building energy modeling, adaptive scaffolding might provide novices with detailed explanations of energy balance equations, heat transfer mechanisms, and HVAC control logic, while providing experts with high-level prompts like "Optimize HVAC control strategy for 20% energy reduction while maintaining thermal comfort" [3]. The LLM adapts its response detail based on student expertise, assessed through prior interactions and performance metrics [3]. The curriculum implements adaptive scaffolding implicitly through its three-phase progression: Months 1--4 provide detailed domain instruction, Months 5--8 introduce LLM tools with structured scaffolding, and Months 9--12 expect students to apply tools independently with minimal guidance. This progression ensures students develop deep expertise before relying on LLM assistance, preventing over-reliance and maintaining critical thinking.

5.3 Automated Error Debugging and HITL Workflows

Automated error debugging is a critical application of LLMs for engineering education, addressing the challenge that novice engineers often struggle to interpret cryptic error messages and identify root causes [7]. LLMs can interpret error messages, suggest corrections, and explain underlying issues, accelerating the debugging process and providing pedagogical value through explanations [43]. **Automated Debugging: Iterative Compilation Feedback Loops** Wan et al. demonstrated automated debugging for Modelica code generation, where LLMs iteratively compile code, interpret error messages, generate corrections, and recompile until successful [43]. This iterative loop continues for up to 5 iterations, after which human intervention is requested if errors persist [43]. The approach reduced Modelica development time by 40--60%, from 10--20 hours to 4--6 hours per module [43]. The key innovation is that LLMs can interpret domain-specific error

messages that are opaque to novices. For example, a Modelica compiler error "Equation count mismatch: 15 equations, 14 unknowns" might confuse a novice, but an LLM can explain: "You have more equations than unknowns, indicating an over-constrained system. Check for duplicate equations or remove one constraint." [43] This pedagogical feedback helps students understand not only how to fix the error but why it occurred [43]. Fu et al. DebugTA system provides LLM-based debugging assistance for programming education, simplifying debugging and teaching through automated breakpoint setting, interactive debugging, and chatting [7]. Participants rated automated breakpoint setting as the most effective feature (5/8 agreement), followed by interactive debugging and chatting (4/8) [7]. All eight participants solved two programming tasks within the given time, demonstrating that LLM-assisted debugging can improve task completion rates [7].

HITL Approval Workflows: 88% Grading Time Reduction, 733% Productivity Increase

Human-in-the-loop (HITL) workflows integrate human oversight at critical decision points, ensuring AI systems do not make consequential decisions autonomously.

For building energy modeling, HITL workflows might require human approval for:

- Energy consumption predictions differing from measured data by >10%
- Retrofit recommendations with cost >\$100,000 or payback period >10 years
- Control strategies that could cause equipment damage or occupant discomfort
- Model calibration when confidence metrics fall below 75%

The hybrid instructor-AI assessment model demonstrated by the literature achieved 88% reduction in grading time and 733% increase in productivity [1]. Feedback quality improved with 100% rubric coverage and 150% increase in anchoring comments to textual evidence [1]. The system demonstrated high reliability ($r = 0.96$) and no bias related to report length, enhancing fairness and quality [1]. These metrics validate the curriculum's emphasis on HITL workflows throughout Months 5--12, teaching students to design approval thresholds, implement audit trails, and integrate human oversight for critical decisions. Students learn that fully autonomous systems often fail in edge cases (where training data is sparse or scenarios are unprecedented), while purely manual processes do not scale. HITL workflows balance automation efficiency with human expertise, achiev-

ing both productivity gains and quality assurance. **Human Evaluation Superiority in Edge Cases** Multiple studies demonstrate that human evaluation outperforms AI evaluation for complex, ambiguous, or novel scenarios. For building energy modeling, edge cases include:

- Buildings with unusual geometries or mixed-use occupancy
- HVAC systems with custom controls or non-standard equipment
- Retrofit scenarios with multiple interacting measures (envelope + HVAC + controls)
- Calibration when measured data exhibits anomalies or sensor failures

In these cases, human experts can apply domain knowledge, recognize patterns from experience, and make judgment calls that AI systems cannot. The curriculum's HITL workflows ensure students learn to identify edge cases, escalate to human review, and document decision rationale for continuous improvement. **Approval Thresholds and Decision Logging** The curriculum teaches students to design approval thresholds based on risk assessment: high-risk decisions (large cost, safety implications, irreversible consequences) require human approval, while low-risk decisions can be automated. For example:

- Energy predictions: approve if error <5%, review if 5--10%, reject if >10%
- Cost estimates: approve if <\$10,000, review if \$10,000--\$100,000, reject if >\$100,000
- Confidence metrics: approve if >90%, review if 75--90%, reject if <75%

Decision logging records all AI decisions, human approvals/rejections, and rationale, creating an audit trail for continuous improvement. Students analyze decision logs to identify patterns: which scenarios frequently require human intervention, which approval thresholds are too conservative or too permissive, and how to refine AI models to reduce false positives/negatives.

5.4 Curriculum Alignment for Section 5

The curriculum's training-first pedagogical model with domain-first progression is strongly validated by the empirical evidence in Section 5. The 38.9% learning improvement for

structured scaffolding [3], 40--60% time savings for LLM-assisted development [43], 88% grading time reduction for HITL workflows [1], and 30--50% data efficiency for physics-constrained models [51] demonstrate that the curriculum's design choices are empirically supported. The three-phase architecture (Domain-First -> AI Integration -> Production Systems) ensures students master building physics before applying AI, preventing spurious correlations and enabling effective physics constraint embedding. The LLM scaffolding approach (structured prompt engineering, LoRA fine-tuning, RAG for normative knowledge) provides powerful productivity tools while maintaining critical thinking and domain expertise. The HITL workflows balance automation efficiency with human oversight, achieving both productivity gains and quality assurance. However, Section 5.2 identified that adaptive scaffolding---where LLM assistance adapts to student expertise---is not explicitly implemented in the current curriculum [3]. The curriculum's three-phase progression provides implicit adaptation (detailed instruction in Months 1--4, structured scaffolding in Months 5--8, independent application in Months 9--12), but explicit adaptive scaffolding could further improve learning outcomes. **Recommended Enhancement:** Add adaptive scaffolding mechanisms to the LLM-based tools used in Months 5--12, where the system assesses student expertise through prior interactions and performance metrics, then adapts response detail accordingly. Novices receive detailed explanations of physical principles and step-by-step guidance, while experts receive high-level hints that preserve cognitive challenge. This enhancement would leverage Cohn et al. theory of adaptive scaffolding [3] while maintaining the curriculum's core strength: domain-first progression that ensures deep expertise before LLM reliance.

6. Conclusion & Curriculum Enhancements

This comprehensive literature review validates the Scientific AI Engineering Master Curriculum through systematic analysis of 888 unique papers published between 2024--2026. The review addressed four critical validation questions, demonstrating strong empirical support for the curriculum's design choices across all domains.

6.1 Validation of the Four Curriculum Pillars

Pillar 1: Advanced PIML for Internal Building Physics Physics-Informed Graph Neural Networks provide empirically validated methods for modeling multi-zone building thermal dynamics, achieving 17--35% accuracy improvements over standard MLPs with R^2 values of 0.79--0.94 [36], [20]. The zones-as-nodes architecture with adjacency matrix encoding explicitly represents building topology, while physics constraints (energy balance, Fourier conduction, thermodynamic laws) are embedded via soft constraints (loss function penalties) and hard constraints (architectural design) [45], [21]. Data efficiency is remarkable: PI-GNNs achieve <1% error using only 20 training samples when physics constraints are embedded, compared to nearly $10\times$ worse performance for pure data-driven GCNs [30]. However, HVAC network modeling as graph elements remains an emerging frontier with critical gaps: limited component coverage (only VAV terminals and ductwork), lack of thermodynamic detail (refrigeration cycles, psychrometrics, fluid flow), no fault detection applications, and scalability concerns. These gaps represent opportunities for curriculum enhancement through explicit HVAC thermodynamics modules and physics-constrained message passing for HVAC networks.

Pillar 2: AI Agents & Simulation Interoperability LLM-based multi-agent systems demonstrate transformative automation capabilities for simulation interoperability. IFC parsing with LLMs achieves 85--92% extraction accuracy with 40--60% time savings, though hallucination risks of 8--15% without guardrails require physics validation [16]. Automated building energy modeling reduces time by >90% (48 minutes vs. 8--32 hours) while achieving ASHRAE Guideline 14-level calibration accuracy (NMBE=2.91%, CV-RMSE=0.139, $R^2=0.972$) [28]. Modelica code generation achieves 100% success for basic logic and 83% for control modules, with 40--60% time savings [43]. Critical challenges remain: semantic gaps between standards (IFC, gbXML, Modelica, FMI), schema mismatches requiring geometric processing, and validation requirements to ensure translations preserve physical consistency. The curriculum's 5-layer safety framework addresses these challenges through Type Validation (schema enforcement), Physics Validation (thermodynamic bounds), and Audit Logging (traceability).

Pillar 3: Multi-Agent Orchestration & Constrained Safety

Generator-optimizer-validator architectures demonstrate substantial performance gains: 90--95% modeling time reduction, 12--23% energy cost savings, and 100% success rates for automated IDF generation [47], [28]. Federated learning enables multi-site portfolio optimization with 12% performance gain and approximately 10% privacy-utility degradation, achieving 23.4% peak demand reduction and 18.7% energy cost reduction [41], [33]. The 5-layer physics-constrained safety framework achieves 95--100% constraint satisfaction rates, reduced rule/physics violation events under constrained workflows, 98.9--100% feasibility for generative design, and 96.67--100% hallucination detection accuracy [38], [39], [47]. These metrics validate that the framework effectively prevents physical hallucinations and catastrophic control violations, supporting the curriculum's emphasis on physics-aware AI.

Pillar 4: Pedagogical Frameworks The training-first pedagogical model with domain-first progression is empirically supported by 38.9% learning improvement for structured scaffolding [3], 60% time savings for LLM-assisted development [43], 88% grading time reduction for HITL workflows [1], and 30--50% data efficiency for physics-constrained models [51]. The three-phase architecture (Domain-First -> AI Integration -> Production Systems) ensures students master building physics before applying AI, preventing spurious correlations and enabling effective physics constraint embedding. LLM scaffolding (structured prompt engineering, LoRA fine-tuning, RAG for normative knowledge) provides powerful productivity tools while maintaining critical thinking and domain expertise. HITL workflows balance automation efficiency with human oversight, achieving both productivity gains and quality assurance.

6.2 Five Specific Curriculum Enhancement Recommendations

Based on the identified gaps and opportunities, we propose five specific curriculum enhancements with module placements: **Enhancement 1: Module 3B -- Intra-Building Graph Construction from IFC Topology (2 weeks)** **Placement:** Month 3 (Building Physics Foundations), after existing EnergyPlus automation content **Content:** Teach students to parse IFC models with LLM agents, extract spatial relationships (IfcSpace, IfcWall, IfcRelSpaceBoundary), and automatically construct zones-as-nodes and walls-

as-edges graphs for PI-GNN thermal modeling. Students implement the IFC-to-Graph translation pipeline, validate extracted topology against building drawings, and handle incomplete IFC models through inference and human-in-the-loop validation. **Learning Outcomes:**

- Parse IFC files using IfcOpenShell and LLM-based semantic querying
- Extract thermal zone geometry, envelope properties, and spatial adjacencies
- Construct adjacency matrices encoding inter-zone heat flow pathways
- Validate IFC-derived graphs against physical constraints (energy balance, connectivity)
- Implement hallucination detection for missing or inconsistent IFC properties

Rationale: Addresses the gap identified in Section 3.1 that IFC-to-Graph automated translation remains largely conceptual with no empirical validation. This module provides hands-on experience with the emerging synergy between LLM-based IFC parsing and PI-GNN thermal modeling, preparing students for end-to-end automation workflows.

Enhancement 2: Module 5A -- Physics-Constrained Message Passing for Zone-to-Zone Heat Transfer (2 weeks) **Placement:** Month 5 (Prompt Engineering & LLM Orchestration), integrated with existing graph neural network content

Content: Teach students to implement physics-constrained message passing in PI-GNNs, where messages propagated along edges represent heat flows governed by Fourier conduction ($Q_{cond} = kA\Delta T/(L\Delta t)$) and convection ($Q_{conv} = hA\Delta T$). Students embed energy balance constraints ($Q_{ext} + Q_1 + Q_2 + Q_3 + Q_4 = Mc\Delta T/\Delta t$) as hard architectural constraints, ensuring predictions satisfy thermodynamic laws by design rather than through loss function penalties. **Learning Outcomes:**

- Implement message-passing neural networks with physics-informed aggregation functions
- Embed Fourier conduction and convection equations as hard architectural constraints
- Design loss functions combining data fitting (L_{data}) and physics compliance

(L_physics)

- Validate PI-GNN predictions against energy balance and thermodynamic feasibility
- Compare soft constraints (loss penalties) vs. hard constraints (architectural design)

Rationale: Addresses the gap identified in Section 2.2 that HVAC network modeling lacks thermodynamic detail and explicit physics constraints. This module teaches students to embed domain-specific physics laws in GNN architectures, leveraging the data efficiency and generalization advantages demonstrated in Section 2.1 (17--35% accuracy improvements, <1% error with 20 samples). **Enhancement 3: Module 7C -- LLM-to-Modelica/FMU Pipeline with Hallucination Guardrails (2 weeks)** **Placement:** Month 7 (Physics Compliance & Anti-Hallucination), integrated with existing safety framework content **Content:** Teach students to use LLMs for Modelica code generation, implement automated debugging loops (iterative compilation, error interpretation, correction), and apply the 5-layer safety framework to catch RAG mismatches and functional errors. Students validate generated code against physical principles (thermodynamic laws, mass conservation, control logic correctness) and functional requirements (equipment protection, occupant comfort, energy efficiency). **Learning Outcomes:**

- Generate Modelica HVAC control modules from natural language specifications
- Implement automated debugging loops with iterative compilation and error correction
- Apply Type Validation (Pydantic schemas) and Physics Validation (thermodynamic bounds)
- Detect RAG mismatches (e.g., And retrieved as Or) through functional testing
- Validate control strategies against equipment protection and comfort constraints

Rationale: Addresses the critical limitation identified in Section 3.4 that RAG often produces mismatches in module selection, and the gap in Section 3.3 that FMI/FMU orchestration with LLMs remains an emerging frontier. This module teaches students to leverage LLM automation (40--60% time savings) while maintaining rigorous validation to prevent hallucinations (8--15% risk without guardrails). **Enhancement 4: Module 9A**

-- gbXML Validation within 5-Layer Safety Framework (1 week) Placement:

Month 9 (Production Deployment), integrated with existing containerization and CI/CD content **Content:** Teach students to validate gbXML exports against thermodynamic constraints (energy balance, U-value ranges, HVAC capacity consistency), detect geometric simplification errors (surface area discrepancies, window-to-wall ratio mismatches), and implement automated surface matching algorithms for inter-zone heat transfer. Students integrate gbXML validation into CI/CD pipelines, ensuring every model commit passes physics checks before deployment. **Learning Outcomes:**

- Validate gbXML files against thermodynamic bounds and building physics constraints
- Detect geometric simplification errors through comparison with source BIM models
- Implement automated surface matching algorithms for adjacent zones
- Integrate gbXML validation into CI/CD pipelines with automated testing
- Generate validation reports documenting constraint satisfaction and identified issues

Rationale: Addresses the gap identified in Section 3.2 that gbXML automation lacks empirical validation of BIM -> gbXML -> EnergyPlus pipelines, and the challenges of geometric simplification and surface matching. This module prepares students to deploy validated building energy models to production systems, leveraging the >90% time reduction demonstrated by Lu et al. while maintaining ASHRAE Guideline 14-level accuracy.

Enhancement 5: Module 11B -- End-to-End IFC -> GNN -> FMU Capstone

(1 week) Placement: Month 11 (Advanced Analytics & Explainability), as a capstone integration exercise before Month 12 **Content:** Students implement a complete end-to-end pipeline: parse IFC models with LLM agents, construct PI-GNN graphs from IFC topology, generate Modelica HVAC controls, orchestrate FMI co-simulation, and validate results against measured data. The capstone synthesizes all curriculum components: domain physics, PI-GNNs, LLM orchestration, multi-agent systems, safety frameworks, and production deployment. **Learning Outcomes:**

- Implement end-to-end IFC -> PI-GNN -> Modelica/FMU -> validation pipeline

- Orchestrate multi-agent workflows (IFC parser, graph constructor, Modelica generator, FMU coordinator)
- Apply 5-layer safety framework at each pipeline stage (Type, Physics, Resource, Audit, Compliance)
- Validate simulation results against measured building operational data
- Generate comprehensive reports documenting pipeline performance, accuracy, and limitations

Rationale: Addresses the gap identified in Section 3.5 that end-to-end IFC -> gbXML -> Modelica/FMU pipelines face significant challenges (semantic gaps, schema mismatches, validation requirements). This capstone provides hands-on experience with the complete workflow, preparing students to bridge the gap between cutting-edge research and industry practice while delivering substantial productivity gains (40--90% time savings) with rigorous validation.

6.3 Top 3 Priority Papers for Curriculum Reading Lists

Based on empirical impact, methodological rigor, and curriculum alignment, we recommend three priority papers for curriculum reading lists:

1. **Jiang, Y., & Dong, B. (2024). Modularized neural network incorporating physical priors for future building energy modeling.** This paper demonstrates the ModNN architecture achieving $R^2 = 0.79\text{--}0.94$ with $\text{MAE} = 0.11\text{--}0.73$ kW, providing a concrete example of zones-as-nodes topology with adjacency matrix encoding and physics constraint embedding [20]. Students should study this paper in Month 4 (PIML) to understand how to design PI-GNN architectures that enforce energy balance and Fourier conduction.
2. **Lu, Y., et al. (2025). Automated building energy modeling for energy retrofits using a large language model-based multi-agent framework. iScience.** This paper demonstrates the Data2BEM framework achieving ASHRAE Guideline 14-level calibration (NMBE=2.91%, CV-RMSE=0.139, $R^2=0.972$) with

>90% time reduction, providing a comprehensive example of generator-optimizer-validator architecture with HITL workflows [28]. Students should study this paper in Month 9 (Production Deployment) to understand how to design multi-agent systems that balance automation efficiency with human oversight.

3. **Sisk, C., et al. (2025). Physics-constrained generative artificial intelligence for rapid takeoff trajectory design.** This paper demonstrates the PhysicsGAN framework achieving 99.6% accuracy, $200\times$ speedup, 98.9% constraint satisfaction, and 100% feasibility, providing empirical validation of physics-constrained safety frameworks [38]. Students should study this paper in Month 7 (Physics Compliance) to understand how to enforce physical constraints in generative AI and prevent hallucinations.

6.4 Future Research Directions

The literature review identified several promising future research directions:

1. **HVAC Network Modeling as Graph Elements:** Extend PI-GNN architectures to model chillers, AHUs, VAV systems, and ductwork as graph nodes with explicit thermodynamic constraints (refrigeration cycles, psychometrics, fluid flow). Validate against real building operational data and demonstrate fault detection capabilities.
2. **Adaptive Scaffolding for LLM-Based Pedagogical Agents:** Develop adaptive scaffolding mechanisms that assess student expertise and adjust LLM response detail accordingly, balancing cognitive challenge with support. Validate through controlled studies measuring learning outcomes and transfer to novel problems.
3. **End-to-End IFC $\rightarrow$ PI-GNN $\rightarrow$ FMU Pipelines:** Implement and validate complete automation pipelines from architect-provided IFC models to calibrated co-simulation models, addressing semantic gaps, schema mismatches, and validation requirements. Quantify accuracy, time savings, and robustness across diverse building types.

4. **Federated PI-GNN Training for Building Portfolios:** Develop federated learning algorithms for PI-GNNs that leverage physics constraints to improve data efficiency and generalization. Quantify privacy-utility tradeoffs and demonstrate performance gains for multi-site building optimization.
5. **Neuro-Symbolic Methods for Equation Discovery:** Extend Kolmogorov-Arnold Networks to discover governing equations for building thermal dynamics from operational data, enabling interpretable physics-informed models. Address training stability, computational efficiency, and scalability challenges.

These research directions align with the curriculum's emphasis on physics-informed AI, multi-agent orchestration, and rigorous safety validation, preparing students to contribute to cutting-edge research while maintaining practical relevance for industry applications.

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